CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI agents with tool calling, MCP and RAG, plus prompting and evals for LLM applications, built as Python backend services. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by hands-on work with FastAPI, Docker and cloud deployment on AWS and Azure.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed and deployed Expert Interface, a graph curation web app (React frontend, Python microservices backend, Docker and Kubernetes stack) on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Delivered as an AI-augmented engineer – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, FastAPI, Postgres, Docker, Kubernetes, Azure, MongoDB, Neo4j, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph that routes each request across vector database queries, general inference, and summarization tool calls, built with business stakeholders around their decision-making workflows.
- Developed a Python backend with FastAPI and a React-based front-end, leveraging Postman for API testing.
- Containerized the entire stack with Docker and deployed it to AWS using CDK as Infrastructure as Code, adhering to clean code practices. Version-controlled via GitLab, with CI/CD deployment pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Postman, Docker, AWS, CDK, EC2, Lambda, GitLab, GitLab CI/CD, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built serverless AWS cloud infrastructure with Lambda, SQS, and DynamoDB, defined as code with CDK, for PrismaChat RAG evaluation.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Vector Databases, Azure AI Search, Hugging Face Transformers, PyTorch, AWS (EC2, VPC, Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Agent Eval & Tracing Harness | Python | LangGraph | Langfuse | LLM-as-a-judge | GitHub Actions

- Evals for a tool-calling LangGraph agent over a mocked business API: golden dataset with regression and adversarial cases, deterministic checks first, then an LLM-as-a-judge; every run traced in Langfuse and gated in GitHub Actions.

TECHNICAL SKILLSET

- **Programming:** Python, SQL | **APIs & Backend:** FastAPI, Postman, Postgres | **Version Control:** Git, GitLab, GitLab CI/CD
- **GenAI & Agents:** LangGraph, LangChain, LiteLLM, MCP, tool calling, RAG, GraphRAG, LLM evals (RAGAS), Hugging Face Transformers, QLoRA
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas
- **Databases:** Vector Databases (Qdrant, Pinecone, Azure AI Search), Postgres, MongoDB, Neo4j
- **Cloud & Deployment:** Docker, Kubernetes, AWS (EC2, Lambda, SQS, DynamoDB, VPC, CDK), Azure, Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles

LANGUAGES

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)